

Interactive Healthy Blood: A Pathway Atlas from One Gene Module to a Thousand Donors

Team 29 | Johnny Betmansour · Robin Sayar · Kevali Shah | ECS 163 — Spring 2026

1. Motivations & Objectives

Whole blood is the most-measured human tissue in transcriptomics, yet every available dataset is visualised in isolation: Tabula Sapiens captures it at single-cell resolution, GTEx provides 803 cross-sectional bulk donors, GSE223613 follows 10 donors every three hours across a full day. No unified interactive view exists that lets a user zoom from the molecular to the cohort scale within a single tool.

We propose an interactive atlas with **five biological scales** — one gene cluster, one cell, one sample, one sample over time, many samples — with pathway activity as the lens connecting all five. A pathway like OxPhos is meaningful at every scale: which genes compose it, which compartments those genes occupy, whether it is active in a given cell type, whether it oscillates with time of day, how much cohort variance it drives. Our hero gene is **SDHA** (succinate dehydrogenase subunit A), the only TCA cycle enzyme also embedded in the inner mitochondrial membrane as Complex II, making it a natural bridge across all five scales.

2. Driving Application & Datasets

We use six data sources, each contributing to a distinct scale:

- **Tabula Sapiens** (blood): 85,233 whole-blood cells, 22 cell types, scRNA-seq. Primary expression source for Levels 3–5.
- **MitoCarta 3.0**: 1,136 experimentally validated human mitochondrial proteins with sub-mitochondrial localization and pathway annotations (107 leaf pathways).
- **Reactome** pathway hierarchy: 873 MitoCarta genes mapped to 16 biologically coherent Reactome subgroups used for Level 3½.
- **GSE223613** (Gosch et al. 2023): 10 donors × 8 timepoints (every 3 h), whole-blood RNA-seq; used for circadian clock face prototype.
- **GSE98582** (Wittenbrink et al. 2018): 11 donors × microarray, dense 24 h time series; comparative circadian dataset with dataset-selector support.
- **GO Cellular Component enrichment** (blood proteome, 4,570 proteins): 247 significant CC terms collapsed to 9 compartments; used for Level 4 cell-diagram treemap.

3. Challenges

- **Cross-scale data compatibility.** PeptideAtlas (PSM counts) and Tabula Sapiens (RNA counts) use different units and scales. We transitioned from PSM to RNA counts mid-project; encoding choices (area, colour) had to be redesigned for RNA.
- **Protein localisation ambiguity.** A single protein can appear in multiple compartments simultaneously. Placing proteins spatially would require an arbitrary choice. We use GO Cellular Component enrichment as a statistical summary — area encodes how many proteins are *annotated* to each compartment, not their physical position.
- **Pathway mosaic problem.** A single pathway can be active in multiple subcellular zones. Spatial placement inside an organelle diagram is intractable. The treemap encoding (area $\propto$ total RNA expression) avoids this by making the spatial metaphor explicit.
- **Circadian data access.** GSE223613 and GSE98582 do not provide programmatic access to raw count matrices without large downloads. Both were approximated with biologically grounded sine-wave models for the prototype; real data integration remains an open item.
- **Co-expression from scRNA-seq.** Computing pairwise Pearson correlations across 85,233 cells requires the full expression matrix; only mean-per-cell-type summaries were retained in our pipeline. A dedicated computation step is required.

• **What to show out of everything.** The data supports far more than any screen can legibly hold — 22 cell types × 107 pathways × thousands of genes, before time and cohort axes. The hard problem is principled reduction: which aggregation keeps the biology readable without flattening it. We settled on the 16-group Reactome hierarchy as the mid-granularity lens (>85% of detected signal, still scannable), but are not fully satisfied — it can merge pathways an expert would keep distinct, and the right granularity likely differs by scale. An open design tension.

4. Background & Related Work

Munzner (2014) is our primary design framework: we apply her channel-expressiveness analysis at every level, ensuring position, area, colour, and motion encode appropriate data types. The persistent pathway constellation is inspired by brushing-and-linking — a selection context maintained across views. Biological network tools (STRING, Cytoscape, WGCNA) established the conventions we follow for co-expression networks: force-directed layout, module-coloured nodes, edge width ∝ correlation. Andrews curves (Andrews 1972) encode multivariate data as Fourier series, letting cluster structure be read from visual density. TimeSignature (Wittenbrink 2018) showed 24-hour internal circadian time is predictable from a single blood sample, motivating our circadian clock face at the '1 sample × 24 h' scale.

5. Method

5.1 Design Overview

The visualization is a single self-contained HTML file (CDN-only, no build step) using D3.js v7, Plotly.js, and Cytoscape.js. Navigation is via a fixed left sidebar with scroll-linked section highlighting using IntersectionObserver. The organising principle (Fig. 1) is a **drill-down** from molecular to cohort scale: the glyph changes at every scale, but a single persistent pathway token keeps the same biology in view throughout.

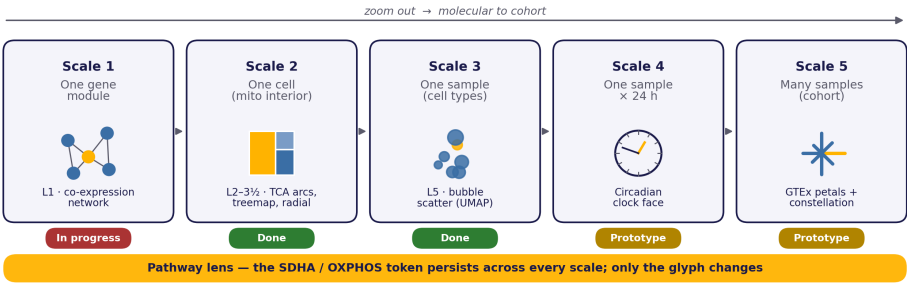


Figure 1. The five biological scales (proposal framing) mapped to the six implemented levels and their glyphs, with build status. The amber pathway lens spans all scales: the SDHA / OXPHOS token persists while only the glyph changes.

5.2 Five Scales, Six Current Levels

Level	Scale	Glyph	Dataset	Status
1	Genome → Gene	Co-expression network	Tabula Sapiens scRNA-seq	In progress
2	Gene → Pathway	TCA cycle arcs (CI bands)	MitoCarta × Tabula Sap.	✓ Done
3	Mito interior	Treemap / Radial petal (toggle)	MitoCarta × Tabula Sap.	✓ Done
3½	Mito Reactome	Treemap (16 categories)	Reactome × Tabula Sap.	✓ Done
4	Cell compartments	Treemap (GO CC enrichment)	GO CC × Tabula Sap.	✓ Done

Level	Scale	Glyph	Dataset	Status
5	Cell types	Bubble scatter (UMAP-style)	Tabula Sapiens	✓ Done

5.3 Encoding Decisions

- **Area $\propto$ total RNA counts** (sum of mean counts across pathway genes): used at Levels 3, 3½, 4. Eliminates p-value encoding; size channel is purely data-driven.
- **Amber = SDHA / OXPHOS hero**: colour is consistent across all levels.
- **No spatial placement of proteins**: GO CC enrichment is a statistical summary, not a physical map. Justified by multi-compartment annotation and dynamic localisation.
- **Toggle between grid and radial petal** at Level 3: treemap for comparison of absolute magnitudes; radial for quick angular overview of 107 pathway petals.

5.4 Standalone Prototypes (pending integration)

- **Constellation panel**: 20 pathway tokens, activity-encoded amber fill, confidence-encoded ring thickness; scores update on scroll via IntersectionObserver; tokens are pinnable.
- **Radial petal glyph**: 20 pathway petals per GTEx donor, petal length $\propto$ |z-score|, amber = over-enriched, blue = under. Uses live GTEx Portal API data.
- **Circadian clock face**: 24 h animated clock with 5 pathway petals; dataset switching (GSE223613 vs GSE98582); individual-donor or mean selection.
- **Andrews curves**: 17 blood cell types mapped via top-5 PCs (85.1% variance); real scRNA-seq PCA, cells coloured by lineage.
- **Co-expression network**: OxPhos module force-directed D3 network, real Pearson correlations from Tabula Sapiens scRNA-seq (in progress).

5.5 Storytelling & Narrative Structure

We adopt a **drill-down** structure rather than a linear martini-glass: the five scales are zoom levels on the same biology, serving both the curious novice (free zooming) and the analyst with a specific question (jump straight to the cohort scale).

- **Persistent pathway constellation as anchor**. A token panel stays beside the canvas at every scale; selecting the SDHA / OXPHOS token keeps it highlighted across scale changes so the reader follows one pathway from gene module to cohort — the constellation is the lens, the canvas is the specimen.
- **Guided entry, then free exploration**. Each level opens with a short scripted intro (e.g. “37 mtDNA genes → edges are co-expression → they rise and fall together”) before handing control to the user.
- **Demo path**. SDHA gene module → mitochondrial interior → cell-type composition → 24-hour oscillation → cohort variance, with the pathway reading aligned at every step.

5.6 Changes & Design Decisions Since Proposal

- **Hero switched to a concrete gene (SDHA)**. The proposal led with abstract pathway tokens (interferon- α , OxPhos); we anchored the narrative to SDHA, the only TCA enzyme also in the inner-membrane Complex II, giving a single thread that is biologically meaningful at all five scales.
- **Expression units: PSM → RNA counts**. We moved from PeptideAtlas spectral counts to Tabula Sapiens scRNA-seq, unifying Levels 3–5 on one measurement type; area/colour encodings were redesigned accordingly.
- **Cell-anatomy diagram → GO CC treemap**. The proposal placed each gene in its primary compartment; multi-compartment and dynamic localisation made spatial placement arbitrary, so Level 4 now uses GO CC enrichment as an explicit statistical summary (area $\propto$ annotated proteins) rather than a fabricated physical map.
- **Added Level 3½ (Reactome grouping)** — a 16-category layer between the 107-leaf MitoCarta treemap and the cell view for a legible mid-granularity overview.

- **Dropped p-value encoding** — size now encodes total RNA expression only, keeping the area channel purely data-driven.
- **Datasets adjusted** — GSE313156 dropped; GSE98582 added as a dense comparative circadian series with a dataset selector.

6. Evaluation Plan

- **Cognitive walkthrough**: can a biology graduate student who did not build the tool follow the SDHA narrative from Level 1 to Level 5 without instruction? Think-aloud, 3–5 participants.
- **Expert validation**: do pathway size differences between neutrophils and monocytes at Level 3 match known biology (monocytes are more metabolically active)?
- **Visual encoding audit (Munzner framework)**: per channel, verify expressiveness (right data type) and effectiveness (most salient available channel).
- **Scalability test**: does the radial petal stay legible at 107 pathways vs 16 Reactome categories? Compare readability at both granularities.

7. Preliminary Results

- **PC1 (59% variance in blood MitoCarta expression) separates progenitors from mature cells**, not immune vs non-immune lineages — hematopoietic stem cells (+59) and erythroid progenitors (+53) highest, erythrocytes and neutrophils (−34) lowest. The dominant axis of mitochondrial variation is differentiation state, not immune activation.
- **Monocytes have ~3× higher total MitoCarta RNA than neutrophils** (279 vs 111 mean counts), visible as markedly larger tiles across all treemap levels. OXPHOS subunits dominate both but are 32% of monocyte vs 27% of neutrophil expression, with monocytes proportionally richer in Translation and Lipid Metabolism.
- **Reactome grouping**: 16 categories describe >85% of detected MitoCarta signal; 107 leaf pathways add granularity to separate TCA cycle from malate-aspartate shuttle.

8. Plan of Activities

Task	Timeline	Status
Levels 2–5 + 3½: six core levels (arcs, treemaps, scatter)	by May 26	✓ Done
Four prototypes (constellation, petal, clock, Andrews)	by May 28	✓ Done
Level 1: co-expression network (scRNA-seq Pearson)	May 29–30	In progress
Integrate constellation + GTEx petal into main viz	May 30–Jun 1	Upcoming
Circadian clock: replace sine models with real data	Jun 1–2	Upcoming
Andrews curves → Level 5 integration	Jun 2–3	Upcoming
Final polish + user evaluation (think-aloud, n=3–5)	Jun 3–4	Upcoming
Final report	Jun 5	Upcoming

9. Team Effort Division

Member	Primary Contributions	Effort (%)
Robin Sayar	Project architecture; all six viz levels (L1–5, 3½); all five prototypes (constellation, radial petal, clock, Andrews curves, co-expression); full data pipeline; report.	~75
Johnny Betmansour	Dataset research (GSE223613, GSE98582, GSE56931); literature review; biological interpretation.	~15
Kevali Shah	Evaluation design; walkthrough protocol; design feedback.	~10

Note: Robin took on most implementation before task assignments were established — a coordination failure, not teammate unwillingness. Explicit ownership has been agreed for all remaining work.

10. References

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